

Subject Name: Cell and Molecular Biology Laboratory

Subject No: BT29204

Prerequisite: Cell and Molecular Biology (BT20204)

Objective:

Laboratory component of its cognate theory subject is very crucial for hands on training to understand the subject. The subject will allow laboratory exposures on various aspects of cell biology and basic concepts of molecular biology, including cell structure, cell division, cell viability, chromosome, cell cytoskeleton, etc. using plant and animal cells. Extraction of genomic and plasmid DNA from bacterial cells, agarose gel electrophoresis of DNA samples, PCR based analysis of genes is included. Restriction enzyme digestion and agarose gel analysis is discussed. Students will be allowed to get an exposure to various high-end instrument related to cellular biology through a visit to our central research facility.

LTP - 0-0-3

Credit - 2

Content:

Experiments (Each experiment of 3 hours duration):

1. Introduction about this laboratory
2. Histochemical study of different plant organs (roots, shoots and leaves)
3. Cell divisions- mitosis & meiosis, exhibiting prepared slides showing different stages
4. Preparation of mitotic cells from onion root tips
5. Preparation of meiotic cells from onion flower buds
6. Fluorescence microscopy of animal cells
7. MTT Assay for cell viability
8. Preparation of E. coli competent cells and plasmid DNA transformation
9. Plasmid DNA isolation, Agarose gel electrophoresis
10. Genomic DNA isolation from animal tissues, Agarose gel electrophoresis
11. Restricted enzyme digestion of plasmid DNA and genomic DNA, Agarose gel electrophoresis
12. Polymerase chain reaction (PCR), Agarose gel electrophoresis